Observation:

Record the observation in the table as follows:

S. No.	Soil types	Weight of soil (X)	Volume of water poured (Y)	Volume of water collected in measuring cylinder (Z)	Volume of water retained by the soil (Y – Z)	Water holding capacity of the soil in percentage (Y – Z) / X × 100
1	Garden soil					
2	Roadside soil					

Inference:

Garden soil has a high water holding capacity than the roadside soil, because roadside soil has larger quantities of sand and silt.

Precautions:

1. Weighing of soil samples should be done accurately.
2. Pour water slowly and gently on the soil in the funnel
3. Record the volume of collected water in the measuring cylinders carefully.

Exercise 23: Isolation of DNA from plant materials

DNA is one of the nucleic acids found in living systems. DNA acts as the genetic material in most of the organisms.

Principle: Recombinant DNA technology has allowed breeders to introduce foreign DNA in other organisms including bacteria, yeast, plants and animals. Such organisms are called Genetically Modified Organisms (GMOs). Thus rDNA technology involves isolation of DNA from a variety of sources and formation of new combination of DNA.

Aim: To isolate DNA from available plant materials such as spinach leaves, fresh green pea seeds, green papaya, etc.

Requirements: Plant materials, mortar and pestle, beakers, test tubes, ethanol, etc.

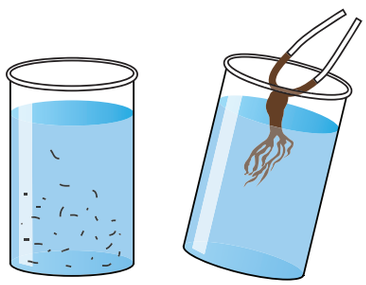


Figure 22: Isolation of DNA

Procedure: Take a small amount of plant material and grind it in a mortar with a little amount of water and sodium chloride. Make it into a solution and filter it. To this filterate, add liquid soap solution or any detergent solution and mix it with a glass rod. Then tilt the test tube and add chilled ethanol and leave it aside in the stand. After half-an-hour we can observe the precipitated DNA as fine threads. DNA that separates can be removed by spooling

Observation: DNA appears as white precipitate of very fine threads on the spool.

Inference: Thus DNA can be isolated from the plant cell nucleus by this technique.

Precautions:

1. All the glasswares must be thoroughly cleaned and dried.
2. The chemicals used for the experiments must be of standard quality.
3. If ordinary ethanol is used, the time duration for obtaining precipitated DNA may extend further.

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